7 Extremal Graph Theory

In this chapter we study how global parameters of a graph, such as its edge density or chromatic number, can influence its local substructures. How many edges, for instance, do we have to give a graph on n vertices to be sure that, no matter how these edges are arranged, the graph will contain a K^r subgraph for some given r? Or at least a K^r minor? Will some sufficiently high average degree or chromatic number ensure that one of these substructures occurs?

Questions of this type are among the most natural ones in graph theory, and there is a host of deep and interesting results. Collectively, these are known as *extremal graph theory*.

Extremal graph problems in this sense fall neatly into two categories, as follows. If we are looking for ways to ensure by global assumptions that a graph G contains some given graph H as a *minor* (or topological minor), it will suffice to raise $\|G\|$ above the value of some linear function of $|G|$, i.e., to make $\varepsilon(G)$ large enough. The precise value of ε needed to force a desired minor or topological minor will be our topic in Section 7.2. Graphs whose number of edges is about[1] linear in their number of vertices are called *sparse*, so Section 7.2 is devoted to 'sparse extremal graph theory'.

sparse

A particularly interesting way to force an H minor is to assume that $\chi(G)$ is large. Recall that if $\chi(G) \geqslant k+1$, say, then G has a subgraph G' with $2\varepsilon(G') \geqslant \delta(G') \geqslant k$ (Lemma 5.2.3). The question here is whether the effect of large χ is limited to this indirect influence via ε, or whether an assumption of $\chi \geqslant k+1$ can force bigger minors than

[1] Formally, the notions of sparse and dense (below) make sense only for classes of graphs whose order tends to infinity, not for individual graphs.

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the assumption of $2\varepsilon \geqslant k$ can. *Hadwiger's conjecture*, which we meet in Section 7.3, asserts that χ has this quality. The conjecture can be viewed as a generalization of the four colour theorem, and is regarded by many as the most challenging open problem in graph theory.

On the other hand, if we ask what global assumptions might imply the existence of some given graph H as a *subgraph*, it will not help to raise invariants such as ε or χ, let alone any of the other invariants discussed in Chapter 1. For as soon as H contains a cycle, there are graphs of arbitrarily large chromatic number not containing H as a subgraph (Theorem 5.2.5). In fact, unless H is bipartite, any function f such that $f(n)$ edges on n vertices force an H subgraph must grow quadratically with n (why?).

dense

Graphs with a number of edges about quadratic in their number of vertices are usually called *dense*; the number $\|G\|/\binom{|G|}{2}$, the proportion of its potential edges that G actually has, is the *edge density* of G. The question of exactly which edge density is needed to force a given subgraph is the archetypal extremal graph problem, and it is our first topic in this chapter (Section 7.1). Rather than attempting to survey the wide field of 'dense extremal graph theory', however, we shall concentrate on its two most important results: we first prove Turán's classical extremal graph theorem for $H = K^r$—a result that has served as a model for countless similar theorems for other graphs H—and then state the fundamental Erdős-Stone theorem, which gives precise asymptotic information for all H at once.

edge density

Although the Erdős-Stone theorem can be proved by elementary means, we shall use the opportunity of its proof to portray a powerful modern proof technique that has transformed much of extremal graph theory in recent years: Szemerédi *regularity lemma*. This lemma is presented and proved in Section 7.4. In Section 7.5, we outline a general method for applying it, and illustrate this in the proof of the Erdős-Stone theorem. Another application of the regularity lemma will be given in Chapter 9.2.

7.1 Subgraphs

Let H be a graph and $n \geqslant |H|$. How many edges will suffice to force an H subgraph in any graph on n vertices, no matter how these edges are arranged? Or, to rephrase the problem: which is the greatest possible number of edges that a graph on n vertices can have *without* containing a copy of H as a subgraph? What will such a graph look like? Will it be unique?

extremal

A graph $G \not\supseteq H$ on n vertices with the largest possible number of edges is called *extremal* for n and H; its number of edges is denoted by

$\mathrm{ex}(n, H)$. Clearly, any graph G that is extremal for some n and H will also be edge-maximal with $H \not\subseteq G$. Conversely, though, edge-maximality does not imply extremality: G may well be edge-maximal with $H \not\subseteq G$ while having fewer than $\mathrm{ex}(n, H)$ edges (Fig. 7.1.1). $\mathrm{ex}(n, H)$



Fig. 7.1.1. Two graphs that are edge-maximal with $P^3 \not\subseteq G$; is the right one extremal?

As a case in point, we consider our problem for $H = K^r$ (with $r > 1$). A moment's thought suggests some obvious candidates for extremality here: all complete $(r-1)$-partite graphs are edge-maximal without containing K^r. But which among these have the greatest number of edges? Clearly those whose partition sets are as equal as possible, i.e. differ in size by at most 1: if V_1, V_2 are two partition sets with $|V_1| - |V_2| \geqslant 2$, we may increase the number of edges in our complete $(r-1)$-partite graph by moving a vertex from V_1 to V_2.

The unique complete $(r-1)$-partite graphs on $n \geqslant r-1$ vertices whose partition sets differ in size by at most 1 are called *Turán graphs*; we denote them by $T^{r-1}(n)$ and their number of edges by $t_{r-1}(n)$ (Fig. 7.1.2). For $n < r-1$ we shall formally continue to use these definitions, with the proviso that—contrary to our usual terminology—the partition sets may now be empty; then, clearly, $T^{r-1}(n) = K^n$ for all $n \leqslant r-1$. $T^{r-1}(n)$ $t_{r-1}(n)$

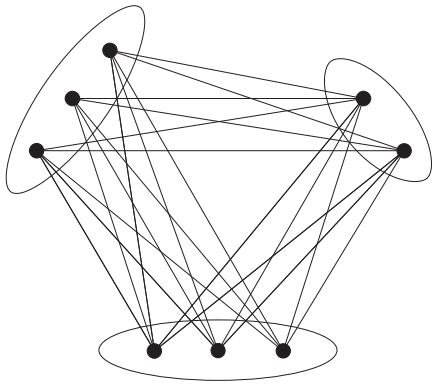


Fig. 7.1.2. The Turán graph $T^3(8)$

The following theorem tells us that $T^{r-1}(n)$ is indeed extremal for n and K^r, and as such unique; in particular, $\mathrm{ex}(n, K^r) = t_{r-1}(n)$.

Theorem 7.1.1. (Turán 1941) [7.1.2] [9.2.2]
For all integers r, n with $r > 1$, every graph $G \not\supseteq K^r$ with n vertices and $\mathrm{ex}(n, K^r)$ edges is a $T^{r-1}(n)$.

We give two proofs: one using induction, the other by a short and direct local argument.

First proof. We apply induction on n. For $n \leqslant r-1$ we have $G = K^n = T^{r-1}(n)$ as claimed. For the induction step, let now $n \geqslant r$.

Since G is edge-maximal without a K^r subgraph, G has a subgraph $K = K^{r-1}$. By the induction hypothesis, $G - K$ has at most $t_{r-1}(n-r+1)$ edges, and each vertex of $G - K$ has at most $r-2$ neighbours in K. Hence,

$$\|G\| \leqslant t_{r-1}(n-r+1) + (n-r+1)(r-2) + \binom{r-1}{2} = t_{r-1}(n)\,; \quad (1)$$

the equality on the right follows by inspection of the Turán graph $T^{r-1}(n)$ (Fig. 7.1.3).

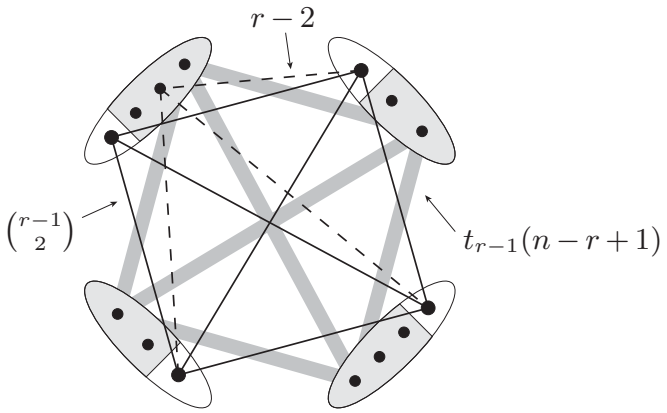


Fig. 7.1.3. The equation from (1) for $r = 5$ and $n = 14$

Since G is extremal for K^r (and $T^{r-1}(n) \not\supseteq K^r$), we have equality in (1). Thus, every vertex of $G - K$ has *exactly* $r-2$ neighbours in K—just like the vertices $x_1, \dots, x_{r-1}$ of K itself. For $i = 1, \dots, r-1$ let

$$V_i := \{\, v \in V(G) \mid vx_i \notin E(G) \,\}$$

be the set of all vertices of G whose $r-2$ neighbours in K are precisely the vertices other than x_i. Since $K^r \not\subseteq G$, each of the sets V_i is independent, and they partition $V(G)$. Hence, G is $(r-1)$-partite. As $T^{r-1}(n)$ is the unique $(r-1)$-partite graph with n vertices and the maximum number of edges, our claim that $G = T^{r-1}(n)$ follows from the assumed extremality of G. □

Written compactly as above, the proof of Turán's theorem may appear a little magical, perhaps even technical. When we look at it more closely, however, we can see how it evolves naturally from the initial idea

of using a subgraph $K \simeq K^{r-1}$ as the seed for the $T^{r-1}(n)$ structure we are hoping to identify in G. Indeed, once we have fixed K and wonder how the rest of G might relate to it, we immediately observe that for every $v \in G-K$ there is a vertex $x \in K$ such that $vx \notin E(G)$. Turning then to the internal structure of $G-K$, we know from the induction hypothesis that the most edges it can possibly have is $t_{r-1}(n-r+1)$, and only if $G-K \simeq T^{r-1}(n-r+1)$. We do not know yet that $G-K$ can indeed have that many edges: all we know is that G, not necessarily $G-K$, has as many edges as possible without a K^r subgraph, and giving $G-K$ the structure of a $T^{r-1}(n-r+1)$ might prevent us from having as many edges between $G-K$ and K as we might otherwise have. But this conflict does not in fact arise. Indeed, we can give $G-K$ this structure *and* have the theoretical maximum number of edges between $G-K$ and K (all except the necessary non-edges of type vx noted earlier) if we form G from K by expanding the $r-1$ vertices of K to the $r-1$ vertex classes of a $T^{r-1}(n)$. And since this is the only way in which both these aims can be achieved, $T^{r-1}(n)$ is once more the unique extremal graph on n vertices without a K^r.

In our second proof of Turán's theorem we shall use an operation called *vertex duplication*. By *duplicating* a vertex $v \in G$ we mean adding to G a new vertex v' and joining it to exactly the neighbours of v (but not to v itself).

vertex duplication

Second proof. We have already seen that among the complete k-partite graphs on n vertices the Turán graphs $T^k(n)$ have the most edges, and their degrees show that $T^{r-1}(n)$ has more edges than any $T^k(n)$ with $k < r-1$. So it suffices to show that G is complete multipartite.

If not, then non-adjacency is not an equivalence relation on $V(G)$, and so there are vertices y_1, x, y_2 such that $y_1x, xy_2 \notin E(G)$ but $y_1y_2 \in E(G)$. If $d(y_1) > d(x)$, then deleting x and duplicating y_1 yields another K^r-free graph with more edges than G, contradicting the choice of G. So $d(y_1) \leqslant d(x)$, and similarly $d(y_2) \leqslant d(x)$. But then deleting both y_1 and y_2 and duplicating x twice yields a K^r-free graph with more edges than G, again contradicting the choice of G. □

The Turán graphs $T^{r-1}(n)$ are dense: in order of magnitude, they have about n^2 edges. More exactly, for every n and r we have

$$t_{r-1}(n) \leqslant \tfrac{1}{2}n^2\,\frac{r-2}{r-1}\,,$$

with equality whenever $r-1$ divides n (Exercise 7). It is therefore remarkable that just ϵn^2 more edges (for any fixed $\epsilon > 0$ and n large) give us not only a K^r subgraph (as does Turán's theorem) but a K^r_s for any given integer s—a graph itself teeming with K^r subgraphs:

Theorem 7.1.2. (Erdős & Stone 1946)
For all integers $r \geqslant 2$ and $s \geqslant 1$, and every $\epsilon > 0$, there exists an integer n_0 such that every graph with $n \geqslant n_0$ vertices and at least

$$t_{r-1}(n) + \epsilon n^2$$

edges contains K_s^r as a subgraph.

A proof of the Erdős-Stone theorem will be given in Section 7.5, as an illustration of how the regularity lemma may be applied. But the theorem can also be proved directly; see the notes for references.

The Erdős-Stone theorem is interesting not only in its own right: it also has a most interesting corollary. In fact, it was this entirely unexpected corollary that established the theorem as a kind of meta-theorem for the extremal theory of dense graphs, and thus made it famous.

Given a graph H and an integer n, consider the number $h_n := \mathrm{ex}(n, H)/\binom{n}{2}$: the maximum edge density that an n-vertex graph can have without containing a copy of H. Could it be that this critical density is essentially just a function of H, that h_n converges as $n \to \infty$? Theorem 7.1.2 implies this, and more: the limit of h_n is determined by a very simple function of a natural invariant of H—its chromatic number!

Corollary 7.1.3. *For every graph H with at least one edge,*

$$\lim_{n\to\infty} \mathrm{ex}(n, H)\binom{n}{2}^{-1} = \frac{\chi(H)-2}{\chi(H)-1}\,.$$

For the proof of Corollary 7.1.3 we need as a lemma that $t_{r-1}(n)$ never deviates much from the value it takes when $r-1$ divides n (see above), and that $t_{r-1}(n)/\binom{n}{2}$ converges accordingly. The proof of the lemma is left as an easy exercise with hint (Exercise 8).

[7.1.2] **Lemma 7.1.4.**

$$\lim_{n\to\infty} t_{r-1}(n)\binom{n}{2}^{-1} = \frac{r-2}{r-1}\,.$$

□

r **Proof of Corollary 7.1.3.** Let $r := \chi(H)$. Since H cannot be coloured with $r-1$ colours, we have $H \not\subseteq T^{r-1}(n)$ for all $n \in \mathbb{N}$, and hence

$$t_{r-1}(n) \leqslant \mathrm{ex}(n, H)\,.$$

On the other hand, $H \subseteq K_s^r$ for all sufficiently large s, so

$$\mathrm{ex}(n, H) \leqslant \mathrm{ex}(n, K_s^r)$$

for all those s. Let us fix such an s. For every $\epsilon > 0$, Theorem 7.1.2 implies that eventually (i.e. for large enough n) s

$$\mathrm{ex}(n, K_s^r) < t_{r-1}(n) + \epsilon n^2.$$

Hence for n large,

$$\begin{aligned} t_{r-1}(n)/\tbinom{n}{2} &\leqslant \mathrm{ex}(n,H)/\tbinom{n}{2} \\ &\leqslant \mathrm{ex}(n,K_s^r)/\tbinom{n}{2} \\ &< t_{r-1}(n)/\tbinom{n}{2} + \epsilon n^2/\tbinom{n}{2} \\ &= t_{r-1}(n)/\tbinom{n}{2} + 2\epsilon/(1-\tfrac{1}{n}) \\ &\leqslant t_{r-1}(n)/\tbinom{n}{2} + 4\epsilon \qquad (\text{assume } n \geqslant 2). \end{aligned}$$

Therefore, since $t_{r-1}(n)/\binom{n}{2}$ converges to $\frac{r-2}{r-1}$ (Lemma 7.1.4), so does $\mathrm{ex}(n,H)/\binom{n}{2}$. □

For bipartite graphs H, Corollary 7.1.3 says that substantially fewer than $\binom{n}{2}$ edges suffice to force an H subgraph. It turns out that

$$c_1 n^{2-\frac{2}{r+1}} \leqslant \mathrm{ex}(n, K_{r,r}) \leqslant c_2 n^{2-\frac{1}{r}}$$

for suitable constants c_1, c_2 depending on r; the lower bound is obtained by random graphs,[2] the upper bound is calculated in Exercise 12. If H is a forest, then $H \subseteq G$ as soon as $\varepsilon(G)$ is large enough, so $\mathrm{ex}(n,H)$ is at most linear in n (Exercise 14). Erdős and Sós conjectured in 1963 that $\mathrm{ex}(n,T) \leqslant \frac{1}{2}(k-1)n$ for all trees with $k \geqslant 2$ edges; as a general bound for all n, this is best possible for every T (Exercises 15–17).

Erdős-Sós conjecture

A related but rather different question is whether large values of ε or χ can force a graph G to contain a given tree T as an *induced* subgraph. Of course, we need some additional assumption for this to make sense—for example, to prevent G from just being a large complete graph. The weakest sensible such assumption is that G has bounded clique number, i.e., that $G \not\supseteq K^r$ for some fixed integer r. Then large average degree still does not force an induced copy of T—consider complete bipartite graphs—but large chromatic number might: according to a remarkable conjecture of Gyárfás (1975), there exists for every $r \in \mathbb{N}$ and every tree T an integer $k = k(T,r)$ such that every graph G with $\chi(G) \geqslant k$ and $\omega(G) < r$ contains T as an induced subgraph.

[2] see Chapter 11

7.2 Minors

In this section and the next we ask to what extent assumptions about invariants of a graph such as average degree, chromatic number, or girth can force it to contain another given graph as a minor or topological minor.

As a starting question, let us consider the analogue of Turán's theorem: how many edges on n vertices force a K^r minor or topological minor? The qualitative answer is that, unlike for K^r subgraphs where we might need as many as $\frac{1}{2}\frac{r-2}{r-1}n^2$ edges, a number of edges linear in n is enough: it suffices to assume that the graph has large enough average degree (depending on r).

Proposition 7.2.1. *Every graph of average degree at least 2^{r-2} has a K^r minor, for all $r \in \mathbb{N}$.*

Proof. We apply induction on r. For $r \leqslant 2$ the assertion is trivial. For the induction step let $r \geqslant 3$, and let G be any graph of average degree at least 2^{r-2}. Then $\varepsilon(G) \geqslant 2^{r-3}$; let H be a minimal minor of G with $\varepsilon(H) \geqslant 2^{r-3}$. Pick a vertex $x \in H$. By the minimality of H, x is not isolated. And each of its neighbours y has at least 2^{r-3} common neighbours with x: otherwise contracting the edge xy would lose us one vertex and at most 2^{r-3} edges, yielding a smaller minor H' with $\varepsilon(H') \geqslant 2^{r-3}$. The subgraph induced in H by the neighbours of x therefore has minimum degree at least 2^{r-3}, and hence has a K^{r-1} minor by the induction hypothesis. Together with x this yields the desired K^r minor of G. □

In Proposition 7.2.1 we needed an average degree of 2^{r-2} to force a K^r minor by induction on r. Forcing a topological K^r minor is a little harder: we shall fix its branch vertices in advance and then construct its subdivided edges inductively, which requires an average degree of $2^{\binom{r}{2}}$ to start with. Apart from this difference, the proof follows the same idea:

Proposition 7.2.2. *Every graph of average degree at least $2^{\binom{r}{2}}$ has a topological K^r minor, for every integer $r \geqslant 2$.*

(1.2.2)
(1.3.1)

Proof. The assertion is clear for $r = 2$, so let us assume that $r \geqslant 3$. We show by induction on $m = r, \dots, \binom{r}{2}$ that every graph G of average degree $d(G) \geqslant 2^m$ has a topological minor X with r vertices and m edges.

If $m = r$ then, by Propositions 1.2.2 and 1.3.1, G contains a cycle of length at least $\varepsilon(G) + 1 \geqslant 2^{r-1} + 1 \geqslant r + 1$, and the assertion follows with $X = C^r$.

Now let $r < m \leqslant \binom{r}{2}$, and assume the assertion holds for smaller m. Let G with $d(G) \geqslant 2^m$ be given; thus, $\varepsilon(G) \geqslant 2^{m-1}$. Since G has a component C with $\varepsilon(C) \geqslant \varepsilon(G)$, we may assume that G is connected.

Consider a maximal set $U \subseteq V(G)$ such that U is connected in G and $\varepsilon(G/U) \geqslant 2^{m-1}$; such a set U exists, because G itself has the form G/U with $|U| = 1$. Since G is connected, we have $N(U) \neq \emptyset$. U

Let $H := G[N(U)]$. If H has a vertex v of degree $d_H(v) < 2^{m-1}$, we may add it to U and obtain a contradiction to the maximality of U: when we contract the edge vv_U in G/U, we lose one vertex and $d_H(v)+1 \leqslant 2^{m-1}$ edges, so ε will still be at least 2^{m-1}. Therefore $d(H) \geqslant \delta(H) \geqslant 2^{m-1}$. By the induction hypothesis, H contains a TY with $|Y| = r$ and $\|Y\| = m-1$. Let x, y be two branch vertices of this TY that are non-adjacent in Y. Since x and y lie in $N(U)$ and U is connected in G, G contains an x–y path whose inner vertices lie in U. Adding this path to the TY, we obtain the desired TX. □ H

In Chapter 3.5 we used the TK^r from Proposition 7.2.2 (stated there as Lemma 3.5.1) for a first proof that large enough connectivity $f(k)$ implies that a graph is k-linked. Later, in Theorem 3.5.3, we saw that connectivity as low as $2k$, coupled with an average degree of at least $16k$, is enough to imply this.

Conversely, we can use the more involved Theorem 3.5.3 to reduce the bound in Proposition 7.2.2 from exponential to quadratic, which is best possible up to a multiplicative constant (Exercise 24):

Theorem 7.2.3. *There is a constant $c \in \mathbb{R}$ such that, for every $r \in \mathbb{N}$, every graph G of average degree $d(G) \geqslant cr^2$ contains K^r as a topological minor.*

Proof. We prove the theorem with $c = 10$. Let G with $d(G) \geqslant 10r^2$ be given. By Theorem 1.4.3 for $k := r^2$, G has a subgraph H with $\kappa(H) \geqslant r^2$ and $\varepsilon(H) > \varepsilon(G) - r^2 \geqslant 4r^2$. For a TK^r in H, pick a set X of r vertices in H as branch vertices, and a set Y of $r(r-1)$ neighbours of X in H, $r-1$ for each vertex in X, as initial subdividing vertices. These are r^2 vertices altogether; they can be chosen distinct, since $\delta(H) \geqslant \kappa(H) \geqslant r^2$. (1.4.3) (3.5.3)

It remains to link up the vertices of Y in pairs, by disjoint paths in $H' := H - X$ corresponding to the edges of K^r. This can be done if Y is linked in H'. We show more generally that H' is $\frac{1}{2}r(r-1)$-linked, by checking that H' satisfies the premise of Theorem 3.5.3 for $k = \frac{1}{2}r(r-1)$. We have $\kappa(H') \geqslant \kappa(H) - r \geqslant r(r-1) = 2k$. And as H' was obtained from H by deleting at most $r|H|$ edges (as well as some vertices), we also have $\varepsilon(H') \geqslant \varepsilon(H) - r \geqslant 4r(r-1) = 8k$. □

For small r one can try to determine the exact number of edges needed to force a TK^r subgraph on n vertices. For $r = 4$, this number is $2n-2$; see Corollary 7.3.2. For $r = 5$, plane triangulations yield a lower bound of $3n-5$ (Corollary 4.2.10). The converse, that $3n-5$ edges

do force a TK^5—not just either a TK^5 or a $TK_{3,3}$, as they do by Corollary 4.2.10 and Kuratowski's theorem—is already a difficult theorem (Mader 1998).

The average degree needed to force an arbitrary K^r minor is less than that for a TK^r, and it is known very precisely; see the notes for the value of c in the following result.

Theorem 7.2.4. (Kostochka 1982)
There exists a constant $c \in \mathbb{R}$ such that, for every $r \in \mathbb{N}$, every graph G of average degree $d(G) \geqslant c\, r\sqrt{\log r}$ contains K^r as a minor. Up to the value of c, this bound is best possible as a function of r.

The easier implication of the theorem, the fact that in general an average degree of $c\, r\sqrt{\log r}$ is needed to force a K^r minor, follows from considering random graphs as introduced in Chapter 11. The converse implication, that this average degree suffices, is proved by methods not dissimilar to the proof of Theorem 3.5.3.

Rather than proving Theorem 7.2.4, therefore, we devote the remainder of this section to another striking aspect of forcing minors: that we can force a K^r minor in a graph simply by raising its girth (as long as we do not merely subdivide edges). At first glance, this may seem almost paradoxical. But it looks more plausible if, rather than trying to force a K^r minor directly, we instead try to force a minor just of large minimum or average degree—which suffices by Theorem 7.2.4. For if the girth g of a graph is large then the ball $\{\, v \mid d(x,v) < \lfloor g/2 \rfloor \,\}$ around a vertex x induces a tree with many leaves, each of which sends all but one of its incident edges away from the tree. Contracting enough disjoint such trees we can thus hope to obtain a minor of large average degree, which in turn will have a large complete minor.

The following lemma realizes this idea.

Lemma 7.2.5. *Let $d, k \in \mathbb{N}$ with $d \geqslant 3$, and let G be a graph of minimum degree $\delta(G) \geqslant d$ and girth $g(G) \geqslant 8k+3$. Then G has a minor H of minimum degree $\delta(H) \geqslant d(d-1)^k$.*

Proof. Let $X \subseteq V(G)$ be maximal with $d(x,y) > 2k$ for all distinct
X $x, y \in X$. For each $x \in X$ put $T_x^0 := \{x\}$. Given $i < 2k$, assume that we have defined disjoint trees $T_x^i \subseteq G$ (one for each $x \in X$) whose vertices together are precisely the vertices at distance at most i from X in G. Joining each vertex at distance $i+1$ from X to a neighbour at distance i, we obtain a similar set of disjoint trees T_x^{i+1}. As every vertex of G has distance at most $2k$ from X (by the maximality of X), the trees
T_x $T_x := T_x^{2k}$ obtained in this way partition the entire vertex set of G. Let H be the minor of G obtained by contracting every T_x.

To prove that $\delta(H) \geqslant d(d-1)^k$, note first that the T_x are induced subgraphs of G, because $\operatorname{diam}(T_x) \leqslant 4k$ and $g(G) > 4k+1$. Similarly, there is at most one edge in G between any two trees T_x and T_y: two such edges, together with the paths joining their ends in T_x and T_y, would form a cycle of length at most $8k+2 < g(G)$. So all the edges leaving T_x are preserved in the contraction.

How many such edges are there? Note that, for every vertex $u \in T_x^{k-1}$, all its $d_G(u) \geqslant d$ neighbours v also lie in T_x: since $d(v,x) \leqslant k$ and $d(x,y) > 2k$ for every other $y \in X$, we have $d(v,y) > k \geqslant d(v,x)$, so v was added to T_x rather than to T_y when those trees were defined. Therefore T_x^k, and hence also T_x, has at least $d(d-1)^{k-1}$ leaves. But every leaf of T_x sends at least $d-1$ edges away from T_x, so T_x sends at least $d(d-1)^k$ edges to (distinct) other trees T_y. □

Lemma 7.2.5 provides Theorem 7.2.4 with the following corollary:

Theorem 7.2.6. (Thomassen 1983)
There exists a function $f\colon \mathbb{N} \to \mathbb{N}$ such that every graph of minimum degree at least 3 and girth at least $f(r)$ has a K^r minor, for all $r \in \mathbb{N}$.

Proof. We prove the theorem with $f(r) := 8\log r + 4\log\log r + c$, for some constant $c \in \mathbb{R}$. Let $k = k(r) \in \mathbb{N}$ be minimal with $3 \cdot 2^k \geqslant c' r\sqrt{\log r}$, where $c' \in \mathbb{R}$ is the constant from Theorem 7.2.4. Then for a suitable constant $c \in \mathbb{R}$ we have $8k+3 \leqslant 8\log r + 4\log\log r + c$, and the result follows by Lemma 7.2.5 and Theorem 7.2.4. □

Large girth can also be used to force a topological K^r minor. We now need some vertices of degree at least $r-1$ to serve as branch vertices, but if we assume a minimum degree of $r-1$ to secure these, we can even get by with a girth bound that is independent of r:

Theorem 7.2.7. (Kühn & Osthus 2002) [7.3.9]
There exists a constant g such that $G \supseteq TK^r$ for every graph G satisfying $\delta(G) \geqslant r-1$ and $g(G) \geqslant g$.

7.3 Hadwiger's conjecture

As we saw in Section 7.2, an average degree of $c\,r\sqrt{\log r}$ suffices to force an arbitrary graph to have a K^r minor, and an average degree of cr^2 forces it to have a topological K^r minor. If we replace 'average degree' above with 'chromatic number' then, with almost the same constants c, the two assertions remain true: this is because every graph with chromatic number k has a subgraph of average degree at least $k-1$ (Lemma 5.2.3).

Although both functions above, $c\,r\sqrt{\log r}$ and cr^2, are best possible (up to the constant c) for the said implications with 'average degree', the question arises whether they are still best possible with 'chromatic number'—or whether some slower-growing function would do in that case. What lies hidden behind this problem about growth rates is a fundamental question about the nature of the invariant χ: can this invariant have some direct *structural* effect on a graph in terms of forcing concrete substructures, or is its effect no greater than that of the 'unstructural' property of having lots of edges somewhere, which it implies trivially?

Neither for general nor for topological minors is the answer to this question known. For general minors, however, the following conjecture of Hadwiger suggests a positive answer:

Conjecture. (Hadwiger 1943)
The following implication holds for every integer $r > 0$ and every graph G:

$$\chi(G) \geqslant r \;\Rightarrow\; G \succcurlyeq K^r.$$

Hadwiger's conjecture is trivial for $r \leqslant 2$, easy for $r = 3$ and $r = 4$ (exercises), and equivalent to the four colour theorem for $r = 5$ and $r = 6$. For $r \geqslant 7$ the conjecture is open, but it is true for line graphs (Exercise 34) and for graphs of large girth (Exercise 32; see also Corollary 7.3.9). Rephrased as $G \succcurlyeq K^{\chi(G)}$, it is true for almost all graphs.[3] In general, the conjecture for $r+1$ implies it for r (exercise).

The Hadwiger conjecture for any fixed r is equivalent to the assertion that every graph without a K^r minor has an $(r-1)$-colouring. In this reformulation, the conjecture raises the question of what the graphs without a K^r minor look like: any sufficiently detailed structural description of those graphs should enable us to decide whether or not they can be $(r-1)$-coloured.

For $r = 3$, for example, the graphs without a K^r minor are precisely the forests (why?), and these are indeed 2-colourable. For $r = 4$, there is also a simple structural characterization of the graphs without a K^r minor:

[12.6.2] **Proposition 7.3.1.** *A graph with at least three vertices is edge-maximal without a K^4 minor if and only if it can be constructed recursively from triangles by pasting*[4] *along K^2s.*

[3] See Chapter 11 for the notion of 'almost all'.

[4] This was defined formally in Chapter 5.5.

Proof. Recall first that every IK^4 contains a TK^4, because $\Delta(K^4) = 3$ (Proposition 1.7.3); the graphs without a K^4 minor thus coincide with those without a topological K^4 minor. The proof that any graph constructible as described is edge-maximal without a K^4 minor is left as an easy exercise; in order to deduce Hadwiger's conjecture for $r = 4$, we only need the converse implication anyhow. We prove this by induction on $|G|$. (1.7.3) (4.4.4)

Let G be given, edge-maximal without a K^4 minor. If $|G| = 3$ then G is itself a triangle, so let $|G| \geqslant 4$ for the induction step. Then G is not complete; let $S \subseteq V(G)$ be a separator of size $\kappa(G)$, and let C_1, C_2 be distinct components of $G - S$. Since S is a minimal separator, every vertex in S has a neighbour in C_1 and another in C_2. If $|S| \geqslant 3$, this implies that G contains three independent paths P_1, P_2, P_3 between a vertex $v_1 \in C_1$ and a vertex $v_2 \in C_2$. Since $\kappa(G) = |S| \geqslant 3$, the graph $G - \{v_1, v_2\}$ is connected and contains a (shortest) path P between two different P_i. Then $P \cup P_1 \cup P_2 \cup P_3$ is a TK^4, a contradiction.

Hence $\kappa(G) \leqslant 2$, and the assertion follows from Lemma 4.4.4[5] and the induction hypothesis. □

One of the interesting consequences of Proposition 7.3.1 is that all the edge-maximal graphs without a K^4 minor have the same number of edges, and are thus all 'extremal':

Corollary 7.3.2. *Every edge-maximal graph G without a K^4 minor has $2|G| - 3$ edges.*

Proof. Induction on $|G|$. □

Corollary 7.3.3. *Hadwiger's conjecture holds for $r = 4$.*

Proof. If G arises from G_1 and G_2 by pasting along a complete graph, then $\chi(G) = \max\{\chi(G_1), \chi(G_2)\}$ (see the proof of Proposition 5.5.2). Hence, Proposition 7.3.1 implies by induction on $|G|$ that all edge-maximal (and hence all) graphs without a K^4 minor can be 3-coloured. □

It is also possible to prove Corollary 7.3.3 by a simple direct argument (Exercise 33).

By the four colour theorem, Hadwiger's conjecture for $r = 5$ follows from the following structure theorem for the graphs without a K^5 minor, just as it follows from Proposition 7.3.1 for $r = 4$. The proof of Theorem 7.3.4 is similar to that of Proposition 7.3.1, but considerably longer. We therefore state the theorem without proof:

[5] The proof of this lemma is elementary and can be read independently of the rest of Chapter 4.

Theorem 7.3.4. (Wagner 1937)
Let G be an edge-maximal graph without a K^5 minor. If $|G| \geqslant 4$ then G can be constructed recursively, by pasting along triangles and K^2s, from plane triangulations and copies of the graph W (Fig. 7.3.1).

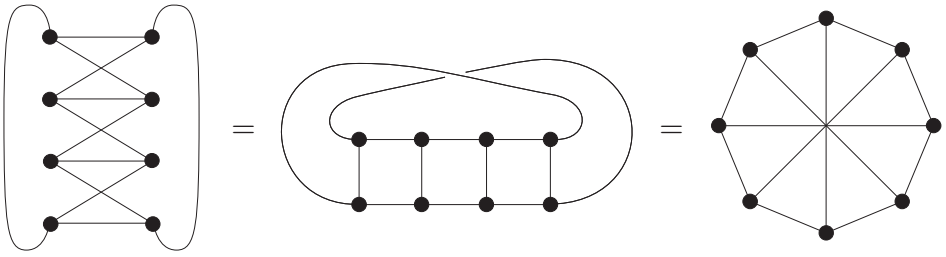


Fig. 7.3.1. Three representations of the *Wagner graph W*

(4.2.10) Using Corollary 4.2.10, one can easily compute which of the graphs constructed as in Theorem 7.3.4 have the most edges. It turns out that these *extremal* graphs without a K^5 minor have no more edges than those that are extremal with respect to $\{IK^5, IK_{3,3}\}$, i.e. the maximal planar graphs:

Corollary 7.3.5. *A graph with n vertices and no K^5 minor has at most $3n-6$ edges.* □

Since $\chi(W) = 3$, Theorem 7.3.4 and the four colour theorem imply Hadwiger's conjecture for $r = 5$:

Corollary 7.3.6. *Hadwiger's conjecture holds for $r = 5$.* □

The Hadwiger conjecture for $r = 6$ is again substantially more difficult than the case $r = 5$, and again it relies on the four colour theorem. The proof shows (without using the four colour theorem) that any minimal-order counterexample arises from a planar graph by adding one vertex—so by the four colour theorem it is not a counterexample after all.

Theorem 7.3.7. (Robertson, Seymour & Thomas 1993)
Hadwiger's conjecture holds for $r = 6$.

As mentioned earlier, the challenge posed by Hadwiger's conjecture is to devise a proof technique that makes better use of the assumption of $\chi \geqslant r$ than just using its consequence of $\delta \geqslant r-1$ in a suitable subgraph, which we know cannot force a K^r minor (Theorem 7.2.4). So far, no such technique is known.

If we resign ourselves to using just $\delta \geqslant r-1$, we can still ask what additional assumptions might help in making this force a K^r minor. Theorem 7.2.7 says that an assumption of large girth has this effect;

see also Exercise 32. In fact, a much weaker assumption suffices: for any fixed $s \in \mathbb{N}$ and all large enough d depending only on s, the graphs $G \not\supseteq K_{s,s}$ of average degree at least d can be shown to have K^r minors for r considerably larger than d. For Hadwiger's conjecture, this implies the following:

Theorem 7.3.8. (Kühn & Osthus 2005)
For every integer s there is an integer r_s such that Hadwiger's conjecture holds for all graphs $G \not\supseteq K_{s,s}$ and $r \geqslant r_s$.

The strengthening of Hadwiger's conjecture that graphs of chromatic number at least r contain K^r as a *topological* minor has become known as *Hajós's conjecture*. It is false in general, but Theorem 7.2.7 implies it for graphs of large girth:

Corollary 7.3.9. *There is a constant g such that all graphs G of girth at least g satisfy the implication $\chi(G) \geqslant r \Rightarrow G \supseteq TK^r$ for all r.*

Proof. Let g be the constant from Theorem 7.2.7. If $\chi(G) \geqslant r$ then, by Lemma 5.2.3, G has a subgraph H of minimum degree $\delta(H) \geqslant r-1$. As $g(H) \geqslant g(G) \geqslant g$, Theorem 7.2.7 implies that $G \supseteq H \supseteq TK^r$. □ (5.2.3) (7.2.7)

7.4 Szemerédi's regularity lemma

Some 40 years ago, in the course of the proof of a theorem about arithmetic progressions of integers, Szemerédi developed a graph-theoretical tool that has since come to dominate methods in extremal graph theory like none other: his *regularity lemma*. Very roughly, the lemma says that all graphs can be approximated by random graphs in the following sense: every graph can be partitioned, into a bounded number of equal parts, so that most of its edges run between different parts and the edges between any two parts are distributed fairly uniformly—just as we would expect it if they had been generated at random.

In order to state the regularity lemma precisely, we need some definitions. Let $G = (V, E)$ be a graph, and let $X, Y \subseteq V$ be disjoint. Then we denote by $\|X, Y\|$ the number of X–Y edges of G, and call $\|X, Y\|$

$$d(X, Y) := \frac{\|X, Y\|}{|X|\,|Y|} \qquad d(X, Y)$$

the *density* of the pair (X, Y). (This is a real number between 0 and 1.) *density*
Given some $\epsilon > 0$, we call a pair (A, B) of disjoint sets $A, B \subseteq V$ *ϵ-regular* if all $X \subseteq A$ and $Y \subseteq B$ with *ϵ-regular pair*

$$|X| \geqslant \epsilon\,|A| \quad \text{and} \quad |Y| \geqslant \epsilon\,|B|$$

satisfy

$$\big|d(X,Y) - d(A,B)\big| \leqslant \epsilon\,.$$

The edges in an ϵ-regular pair are thus distributed fairly uniformly, the more so the smaller the ϵ we started with.

exceptional set

Consider a partition $\{V_0, V_1, \dots, V_k\}$ of V in which one set V_0 has been singled out as an *exceptional set*. (This exceptional set V_0 may be empty.[6]) We call such a partition an *ϵ-regular* partition of G if it satisfies the following three conditions:

ϵ-regular partition

(i) $|V_0| \leqslant \epsilon\,|V|$;

(ii) $|V_1| = \ldots = |V_k|$;

(iii) all but at most ϵk^2 of the pairs (V_i, V_j) with $1 \leqslant i < j \leqslant k$ are ϵ-regular.

The role of the exceptional set V_0 is one of pure convenience: it makes it possible to require that all the other partition sets have exactly the same size. Since condition (iii) affects only the sets $V_1, \dots, V_k$, we may think of V_0 as a kind of bin: its vertices are disregarded when the regularity of the partition is assessed, but there are only few such vertices.

[7.1.2] [9.2.2]

Theorem 7.4.1. (Regularity Lemma)
For every $\epsilon > 0$ and every integer $m \geqslant 1$ there exists an integer M such that every graph of order at least m admits an ϵ-regular partition $\{V_0, V_1, \dots, V_k\}$ with $m \leqslant k \leqslant M$.

The regularity lemma thus says that, given any $\epsilon > 0$, every graph has an ϵ-regular partition into a bounded number of sets. The upper bound M on the number of partition sets ensures that for large graphs the partition sets are large too; note that ϵ-regularity is trivial when the partition sets are singletons, and a powerful property when they are large. The lemma also allows us to specify a lower bound m for the number of partition sets. This can be used to increase the proportion of edges running between different partition sets (i.e., of edges governed by the regularity assertion) over edges inside partition sets (about which we know nothing). See Exercise 38 for more details.

Note that the regularity lemma in this form is designed for use with dense graphs:[7] for sparse graphs it becomes trivial, because all densities of pairs—and hence their differences—tend to zero (Exercise 39).

The remainder of this section is devoted to the proof of the regularity lemma. Although the proof is not difficult, a reader meeting the

[6] So V_0 may be an exception also to our terminological rule that partition sets are not normally empty.

[7] Sparse versions were developed later; see the notes.

regularity lemma here for the first time is likely to draw more insight from seeing how the lemma is typically applied than from studying the technicalities of its proof. Any such reader is encouraged to skip to the start of Section 7.5 now and come back to the proof at his or her leisure.

We shall need the following inequality for reals $\mu_1, \dots, \mu_k > 0$ and $e_1, \dots, e_k \geqslant 0$:

$$\sum \frac{e_i^2}{\mu_i} \geqslant \frac{\left(\sum e_i\right)^2}{\sum \mu_i} \,. \tag{1}$$

This follows from the Cauchy-Schwarz inequality $\sum a_i^2 \sum b_i^2 \geqslant (\sum a_i b_i)^2$ by taking $a_i := \sqrt{\mu_i}$ and $b_i := e_i/\sqrt{\mu_i}$.

Let $G = (V, E)$ be a graph and $n := |V|$. For disjoint sets $A, B \subseteq V$ we define $G = (V, E)$ n

$$q(A, B) := \frac{|A|\,|B|}{n^2}\, d^2(A, B) = \frac{\|A, B\|^2}{|A|\,|B|\,n^2} \,. \qquad q(A, B)$$

For partitions $\mathcal{A}$ of A and $\mathcal{B}$ of B we set

$$q(\mathcal{A}, \mathcal{B}) := \sum_{A' \in \mathcal{A};\ B' \in \mathcal{B}} q(A', B') \,, \qquad q(\mathcal{A}, \mathcal{B})$$

and for a partition $\mathcal{P} = \{C_1, \dots, C_k\}$ of V we let

$$q(\mathcal{P}) := \sum_{i<j} q(C_i, C_j) \,. \qquad q(\mathcal{P})$$

However, if $\mathcal{P} = \{C_0, C_1, \dots, C_k\}$ is a partition of V *with exceptional set* C_0, we treat C_0 as a set of singletons and define

$$q(\mathcal{P}) := q(\tilde{\mathcal{P}}) \,,$$

where $\tilde{\mathcal{P}} := \big\{C_1, \dots, C_k\big\} \cup \big\{\{v\} : v \in C_0\big\}$. $\tilde{\mathcal{P}}$

The function $q(\mathcal{P})$ plays a pivotal role in the proof of the regularity lemma. On the one hand, it measures the regularity of the partition $\mathcal{P}$: if $\mathcal{P}$ has too many irregular pairs (A, B), we may take the pairs (X, Y) of subsets violating the regularity of the pairs (A, B) and make those sets X and Y into partition sets of their own; as we shall prove, this refines $\mathcal{P}$ into a partition for which q is substantially greater than for $\mathcal{P}$. Here, 'substantial' means that the increase of $q(\mathcal{P})$ is bounded below by some

constant depending only on ϵ. On the other hand,

$$\begin{aligned} q(\mathcal{P}) &= \sum_{i<j} q(C_i, C_j) \\ &= \sum_{i<j} \frac{|C_i|\,|C_j|}{n^2}\, d^2(C_i, C_j) \\ &\leqslant \frac{1}{n^2} \sum_{i<j} |C_i|\,|C_j| \\ &\leqslant 1\,. \end{aligned}$$

The number of times that $q(\mathcal{P})$ can be increased by a constant is thus also bounded by a constant—in other words, after some bounded number of refinements our partition will be ϵ-regular! To complete the proof of the regularity lemma, all we have to do then is to note how many sets that last partition can possibly have if we start with a partition into m sets, and to choose this number as our promised bound M.

Let us make all this precise. We begin by showing that, when we refine a partition, the value of q will not decrease:

Lemma 7.4.2.

(i) *Let $C, D \subseteq V$ be disjoint. If $\mathcal{C}$ is a partition of C and $\mathcal{D}$ is a partition of D, then $q(\mathcal{C}, \mathcal{D}) \geqslant q(C, D)$.*

(ii) *If $\mathcal{P}, \mathcal{P}'$ are partitions of V and $\mathcal{P}'$ refines $\mathcal{P}$, then $q(\mathcal{P}') \geqslant q(\mathcal{P})$.*

Proof. (i) Let $\mathcal{C} =: \{C_1, \dots, C_k\}$ and $\mathcal{D} =: \{D_1, \dots, D_\ell\}$. Then

$$\begin{aligned} q(\mathcal{C}, \mathcal{D}) \;&=\; \sum_{i,j} q(C_i, D_j) \\ &=\; \frac{1}{n^2} \sum_{i,j} \frac{\|C_i, D_j\|^2}{|C_i|\,|D_j|} \\ &\underset{(1)}{\geqslant}\; \frac{1}{n^2} \frac{\big(\sum_{i,j} \|C_i, D_j\|\big)^2}{\sum_{i,j} |C_i|\,|D_j|} \\ &=\; \frac{1}{n^2} \frac{\|C, D\|^2}{\big(\sum_i |C_i|\big)\big(\sum_j |D_j|\big)} \\ &=\; q(C, D)\,. \end{aligned}$$

(ii) Let $\mathcal{P} =: \{C_1, \dots, C_k\}$, and for $i = 1, \dots, k$ let $\mathcal{C}_i$ be the partition

of C_i induced by $\mathcal{P}'$. Then

$$\begin{aligned} q(\mathcal{P}) &= \sum_{i<j} q(C_i, C_j) \\ &\underset{\text{(i)}}{\leqslant} \sum_{i<j} q(\mathcal{C}_i, \mathcal{C}_j) \\ &\leqslant q(\mathcal{P}')\,, \end{aligned}$$

since $q(\mathcal{P}') = \sum_i q(\mathcal{C}_i) + \sum_{i<j} q(\mathcal{C}_i, \mathcal{C}_j)$. □

Next, we show that refining a partition by subpartitioning an irregular pair of partition sets increases the value of q a little; since we are dealing here with a single pair only, the amount of this increase will still be less than any constant.

Lemma 7.4.3. *Let $\epsilon > 0$, and let $C, D \subseteq V$ be disjoint. If (C, D) is not ϵ-regular, then there are partitions $\mathcal{C} = \{C_1, C_2\}$ of C and $\mathcal{D} = \{D_1, D_2\}$ of D such that*

$$q(\mathcal{C}, \mathcal{D}) \geqslant q(C, D) + \epsilon^4 \frac{|C|\,|D|}{n^2}\,.$$

Proof. Suppose (C, D) is not ϵ-regular. Then there are sets $C_1 \subseteq C$ and $D_1 \subseteq D$ with $|C_1| \geqslant \epsilon\,|C|$ and $|D_1| \geqslant \epsilon\,|D|$ such that

$$|\eta| > \epsilon \tag{2}$$

for $\eta := d(C_1, D_1) - d(C, D)$. Let $\mathcal{C} := \{C_1, C_2\}$ and $\mathcal{D} := \{D_1, D_2\}$, where $C_2 := C \setminus C_1$ and $D_2 := D \setminus D_1$. η

Let us show that $\mathcal{C}$ and $\mathcal{D}$ satisfy the conclusion of the lemma. We shall write $c_i := |C_i|$, $d_i := |D_i|$, $e_{ij} := \|C_i, D_j\|$, $c := |C|$, $d := |D|$ and $e := \|C, D\|$. As in the proof of Lemma 7.4.2, c_i, d_i, e_{ij} c, d, e

$$\begin{aligned} q(\mathcal{C}, \mathcal{D}) &= \frac{1}{n^2} \sum_{i,j} \frac{e_{ij}^2}{c_i d_j} \\ &= \frac{1}{n^2} \left(\frac{e_{11}^2}{c_1 d_1} + \sum_{i+j>2} \frac{e_{ij}^2}{c_i d_j} \right) \\ &\underset{(1)}{\geqslant} \frac{1}{n^2} \left(\frac{e_{11}^2}{c_1 d_1} + \frac{(e - e_{11})^2}{cd - c_1 d_1} \right). \end{aligned}$$

By definition of η, we have $e_{11} = c_1 d_1 e/cd + \eta c_1 d_1$, so

$$\begin{aligned} n^2\, q(\mathcal{C},\mathcal{D}) &\geqslant \frac{1}{c_1 d_1}\left(\frac{c_1 d_1 e}{cd} + \eta c_1 d_1\right)^2 \\ &\qquad + \frac{1}{cd - c_1 d_1}\left(\frac{cd - c_1 d_1}{cd} e - \eta c_1 d_1\right)^2 \\ &= \frac{c_1 d_1 e^2}{c^2 d^2} + \frac{2e\eta c_1 d_1}{cd} + \eta^2 c_1 d_1 \\ &\qquad + \frac{cd - c_1 d_1}{c^2 d^2} e^2 - \frac{2e\eta c_1 d_1}{cd} + \frac{\eta^2 c_1^2 d_1^2}{cd - c_1 d_1} \\ &\geqslant \frac{e^2}{cd} + \eta^2 c_1 d_1 \\ &\underset{(2)}{\geqslant} \frac{e^2}{cd} + \epsilon^4 cd \end{aligned}$$

since $c_1 \geqslant \epsilon c$ and $d_1 \geqslant \epsilon d$ by the choice of C_1 and D_1. □

Finally, we show that if a partition has enough irregular pairs of partition sets to fall short of the definition of an ϵ-regular partition, then subpartitioning all those pairs at once results in an increase of q by a constant:

Lemma 7.4.4. *Let $0 < \epsilon \leqslant 1/4$, and let $\mathcal{P} = \{C_0, C_1, \ldots, C_k\}$ be a partition of V, with exceptional set C_0 of size $|C_0| \leqslant \epsilon n$ and*
c *$|C_1| = \ldots = |C_k| =: c$. If $\mathcal{P}$ is not ϵ-regular, then there is a partition $\mathcal{P}' = \{C_0', C_1', \ldots, C_\ell'\}$ of V with exceptional set C_0', where $k \leqslant \ell \leqslant k4^{k+1}$, such that $|C_0'| \leqslant |C_0| + n/2^k$, all other sets C_i' have equal size, and either $\mathcal{P}'$ is ϵ-regular or*

$$q(\mathcal{P}') \geqslant q(\mathcal{P}) + \epsilon^5/2\,.$$

$\mathcal{C}_{ij}$ *Proof.* For all $1 \leqslant i < j \leqslant k$, let us define a partition $\mathcal{C}_{ij}$ of C_i and a partition $\mathcal{C}_{ji}$ of C_j, as follows. If the pair (C_i, C_j) is ϵ-regular, we let $\mathcal{C}_{ij} := \{C_i\}$ and $\mathcal{C}_{ji} := \{C_j\}$. If not, then by Lemma 7.4.3 there are partitions $\mathcal{C}_{ij}$ of C_i and $\mathcal{C}_{ji}$ of C_j with $|\mathcal{C}_{ij}| = |\mathcal{C}_{ji}| = 2$ and

$$q(\mathcal{C}_{ij}, \mathcal{C}_{ji}) \geqslant q(C_i, C_j) + \epsilon^4 \frac{|C_i|\,|C_j|}{n^2} = q(C_i, C_j) + \frac{\epsilon^4 c^2}{n^2}\,. \tag{3}$$

$\mathcal{C}_i$ For each $i = 1, \ldots, k$, let $\mathcal{C}_i$ be the unique minimal partition of C_i that refines every partition $\mathcal{C}_{ij}$ with $j \neq i$. (In other words, if we consider two elements of C_i as equivalent whenever they lie in the same partition set

of $\mathcal{C}_{ij}$ for every $j \neq i$, then $\mathcal{C}_i$ is the set of equivalence classes.) Thus, $|\mathcal{C}_i| \leqslant 2^{k-1}$. Now consider the partition

$$\mathcal{C} := \{C_0\} \cup \bigcup_{i=1}^{k} \mathcal{C}_i \qquad \mathcal{C}$$

of V, with C_0 as exceptional set. Then $\mathcal{C}$ refines $\mathcal{P}$ and $|\mathcal{C} \setminus \{C_0\}| \leqslant k2^{k-1}$, so

$$k \leqslant |\mathcal{C}| \leqslant k2^k. \tag{4}$$

Let $\mathcal{C}_0 := \big\{\{v\} \mid v \in C_0\big\}$. Now if $\mathcal{P}$ is not ϵ-regular, then for more than ϵk^2 of the pairs (C_i, C_j) with $1 \leqslant i < j \leqslant k$ the partition $\mathcal{C}_{ij}$ is non-trivial. Hence, by our definition of q for partitions with exceptional set, and Lemma 7.4.2 (i), $\mathcal{C}_0$

$$\begin{aligned}
q(\mathcal{C}) &= \sum_{1\leqslant i<j} q(\mathcal{C}_i, \mathcal{C}_j) + \sum_{1\leqslant i} q(\mathcal{C}_0, \mathcal{C}_i) + \sum_{0\leqslant i} q(\mathcal{C}_i) \\
&\geqslant \sum_{1\leqslant i<j} q(\mathcal{C}_{ij}, \mathcal{C}_{ji}) + \sum_{1\leqslant i} q\big(\mathcal{C}_0, \{C_i\}\big) + q(\mathcal{C}_0) \\
&\underset{(3)}{\geqslant} \sum_{1\leqslant i<j} q(C_i, C_j) + \epsilon k^2 \frac{\epsilon^4 c^2}{n^2} + \sum_{1\leqslant i} q\big(\mathcal{C}_0, \{C_i\}\big) + q(\mathcal{C}_0) \\
&= q(\mathcal{P}) + \epsilon^5 \left(\frac{kc}{n}\right)^2 \\
&\geqslant q(\mathcal{P}) + \epsilon^5/2\,.
\end{aligned}$$

(For the last inequality, recall that $|C_0| \leqslant \epsilon n \leqslant \frac{1}{4}n$, so $kc \geqslant \frac{3}{4}n$.)

In order to turn $\mathcal{C}$ into our desired partition $\mathcal{P}'$, all that remains to do is to cut its sets up into pieces of some common size, small enough that all remaining vertices can be collected into the exceptional set without making this too large.

If $c < 4^k$, the ϵ-regular partition $\mathcal{P}'$ into $C_0' := C_0$ and the singletons $\{v\}$ with $v \in V \setminus C_0$ is as desired, since there are ℓ such singletons for $k \leqslant \ell = kc < k4^k$.

Assume now that $c \geqslant 4^k$. Let $C_1', \dots, C_\ell'$ be a maximal collection of disjoint sets of size $d := \lfloor c/4^k \rfloor \geqslant 1$ such that each C_i' is contained in some $C \in \mathcal{C} \setminus \{C_0\}$, and put $C_0' := V \setminus \bigcup C_i'$. Then $\mathcal{P}' = \{C_0', C_1', \dots, C_\ell'\}$ is indeed a partition of V. Moreover, $\tilde{\mathcal{P}}'$ refines $\tilde{\mathcal{C}}$, so d $\mathcal{P}'$

$$q(\mathcal{P}') \geqslant q(\mathcal{C}) \geqslant q(\mathcal{P}) + \epsilon^5/2$$

by Lemma 7.4.2 (ii). Since each set $C_i' \neq C_0'$ is also contained in one of the sets $C_1, \dots, C_k$, but no more than $c/d \leqslant 4^{k+1}$ sets C_i' can lie inside

the same C_j (by the choice of d), we also have $k \leqslant \ell \leqslant k4^{k+1}$ as required. Finally, the sets $C'_1, \ldots, C'_\ell$ use all but at most d vertices from each set $C \neq C_0$ of $\mathcal{C}$. Hence,

$$\begin{aligned} |C'_0| &\leqslant |C_0| + d\,|\mathcal{C}| \\ &\underset{(4)}{\leqslant} |C_0| + \frac{c}{4^k} k 2^k \\ &= |C_0| + ck/2^k \\ &\leqslant |C_0| + n/2^k. \end{aligned}$$

□

The proof of the regularity lemma now follows easily by repeated application of Lemma 7.4.4:

ϵ, m **Proof of Theorem 7.4.1.** Let $\epsilon > 0$ and $m \geqslant 1$ be given, assuming
s without loss of generality that $\epsilon \leqslant 1/4$. Let $s := 2/\epsilon^5$. This number s is an upper bound on the number of iterations of Lemma 7.4.4 that can be applied to a partition of a graph before it becomes ϵ-regular; recall that $q(\mathcal{P}) \leqslant 1$ for all partitions $\mathcal{P}$.

There is one formal requirement which a partition $\{C_0, C_1, \ldots, C_k\}$ with $|C_1| = \ldots = |C_k|$ has to satisfy before Lemma 7.4.4 can be (re-) applied: the size $|C_0|$ of its exceptional set must not exceed ϵn. With each iteration of the lemma, however, the size of the exceptional set can grow by up to $n/2^k$. (More precisely, by up to $n/2^\ell$, where ℓ is the number of other sets in the current partition; but $\ell \geqslant k$ by the lemma, so $n/2^k$ is certainly an upper bound for the increase.) We thus want to start with k large enough that even s increments of $n/2^k$ add up to at most $\frac{1}{2}\epsilon n$, and ensure that n large enough that, for any initial value of $|C_0| < k$, we have $|C_0| \leqslant \frac{1}{2}\epsilon n$. (If we give our starting partition k non-exceptional sets $C_1, \ldots, C_k$, we should allow an initial size of up to k for C_0, to be able to achieve $|C_1| = \ldots = |C_k|$.)

k So let $k \geqslant m$ be large enough that $2^{k-1} \geqslant s/\epsilon$. Then $s/2^k \leqslant \epsilon/2$, and hence

$$k + \frac{s}{2^k} n \leqslant \epsilon n \tag{5}$$

whenever $k/n \leqslant \epsilon/2$, i.e. for all $n \geqslant 2k/\epsilon$.

Let us now choose M. This should be an upper bound on the number of (non-exceptional) sets in our partition after up to s iterations of Lemma 7.4.4, where in each iteration this number may grow from its current value r to at most $r4^{r+1}$. So let f be the function $x \mapsto x4^{x+1}$,
M and take $M := \max\{f^s(k), 2k/\epsilon\}$; the second term in the maximum ensures that any $n \geqslant M$ is large enough to satisfy (5).

We finally have to show that every graph $G = (V, E)$ of order at least m has an ϵ-regular partition $\{V_0, V_1, \ldots, V_{k'}\}$ with $m \leqslant k' \leqslant M$. So

let G be given, and let $n := |G|$. If $n \leqslant M$, we partition G into $k' := n$ singletons, choosing $V_0 := \emptyset$ and $|V_1| = \ldots = |V_{k'}| = 1$. This partition of G is clearly ϵ-regular. Suppose now that $n > M$. Let $C_0 \subseteq V$ be minimal such that our earlier $k =: k'$ divides $|V \setminus C_0|$, and let $\{C_1, \ldots, C_k\}$ be any partition of $V \setminus C_0$ into sets of equal size. Then $|C_0| < k$, and hence $|C_0| \leqslant \epsilon n$ by (5). Starting with $\{C_0, C_1, \ldots, C_k\}$ we apply Lemma 7.4.4 again and again, until the partition of G obtained is ϵ-regular; this will happen after at most s iterations, since by (5) the size of the exceptional set in the partitions stays below ϵn, so the lemma could indeed be reapplied up to the theoretical maximum of s times. □ n

7.5 Applying the regularity lemma

The purpose of this section is to illustrate how the regularity lemma is typically applied in the context of (dense) extremal graph theory. Suppose we are trying to prove that a certain edge density of a graph G suffices to force the occurrence of some given subgraph H, and that we have an ϵ-regular partition of G. For most of the pairs (V_i, V_j) of partition sets, the edges between V_i and V_j are distributed fairly uniformly; their density, however, may depend on the pair. But since G has many edges, this density cannot be too small for too many pairs: some sizeable proportion of the pairs will have at least a certain positive density. Moreover if G is large, then so are the pairs: recall that the number of partition sets is bounded, and they have equal size. But any large enough bipartite graph with equal partition sets, fixed positive edge density (however small) and a uniform distribution of edges will contain any given bipartite subgraph;[8] this will be made precise below. Writing H as a union of bipartite subgraphs, say those induced by pairs of colour classes of some vertex colouring of H, we shall obtain $H \subseteq G$ as desired.

These ideas will be formalized by Lemma 7.5.2 below. We shall then use this and the regularity lemma to prove the Erdős-Stone theorem from Section 7.1; another application will be given later, in the proof of Theorem 9.2.2. We wind up the section with an informal review of the application of the regularity lemma that we have seen, summarizing what it can teach us for similar applications. In particular, we look at how the various parameters involved depend on each other, and in which order they have to be chosen to make the lemma work.

Let us begin by noting a simple consequence of the ϵ-regularity of a pair (A, B). For any subset $Y \subseteq B$ that is not too small, most vertices of A have about the expected number of neighbours in Y:

[8] Readers already acquainted with random graphs may find it instructive to compare this statement with Proposition 11.3.1.

Lemma 7.5.1. *Let (A,B) be an ϵ-regular pair, of density d say, and let $Y \subseteq B$ have size $|Y| \geqslant \epsilon\,|B|$. Then all but fewer than $\epsilon\,|A|$ of the vertices in A have (each) at least $(d-\epsilon)|Y|$ neighbours in Y.*

Proof. Let $X \subseteq A$ be the set of vertices with fewer than $(d-\epsilon)|Y|$ neighbours in Y. Then $\|X,Y\| < |X|(d-\epsilon)|Y|$, so

$$d(X,Y) = \frac{\|X,Y\|}{|X|\,|Y|} < d-\epsilon = d(A,B)-\epsilon\,.$$

As (A,B) is ϵ-regular and $|Y| \geqslant \epsilon\,|B|$, this implies that $|X| < \epsilon\,|A|$. □

R — $\textit{regularity graph}$ — V_i^s — R_s

Let G be a graph with an ϵ-regular partition $\{V_0, V_1, \dots, V_k\}$, with exceptional set V_0 and $|V_1| = \dots = |V_k| =: \ell$. Given $d \in [0,1]$, let R be the graph on $\{V_1,\dots,V_k\}$ in which two vertices V_i, V_j are adjacent if and only if they form an ϵ-regular pair in G of density $\geqslant d$. We shall call R a *regularity graph* of G with parameters ϵ, ℓ and d. Given $s \in \mathbb{N}$, let us now replace every vertex V_i of R by a set V_i^s of s vertices, and every edge by a complete bipartite graph between the corresponding s-sets. The resulting graph will be denoted by R_s. (For $R = K^r$, for example, we have $R_s = K_s^r$.)

The following lemma says that subgraphs of R_s can also be found in G, provided that $d > 0$, that ϵ is small enough, and that the V_i are large enough. In fact, the values of ϵ and ℓ required depend only on (d and) the maximum degree of the subgraph:

[9.2.2] **Lemma 7.5.2.** *For all $d \in (0,1]$ and $\Delta \geqslant 1$ there exists an $\epsilon_0 > 0$ with the following property: if G is any graph, H is a graph with $\Delta(H) \leqslant \Delta$, $s \in \mathbb{N}$, and R is any regularity graph of G with parameters $\epsilon \leqslant \epsilon_0$, $\ell \geqslant 2s/d^\Delta$ and d, then*

$$H \subseteq R_s \;\Rightarrow\; H \subseteq G\,.$$

d, Δ, ϵ_0 — *Proof.* Given d and Δ, choose $\epsilon_0 > 0$ small enough that $\epsilon_0 < d$ and

$$(d-\epsilon_0)^\Delta - \Delta\epsilon_0 \geqslant \tfrac{1}{2}d^\Delta\,; \tag{1}$$

G, H, R, R_s — V_i — ϵ, k, ℓ — u_i, h — σ — v_i

such a choice is possible, since $(d-\epsilon)^\Delta - \Delta\epsilon \to d^\Delta$ as $\epsilon \to 0$. Now let G, H, s and R be given as stated. Let $\{V_0, V_1, \dots, V_k\}$ be the ϵ-regular partition of G that gave rise to R; thus, $\epsilon \leqslant \epsilon_0$, $V(R) = \{V_1,\dots,V_k\}$ and $|V_1| = \dots = |V_k| = \ell \geqslant 2s/d^\Delta$. Let us assume that H is actually a subgraph of R_s (not just isomorphic to one), with vertices $u_1,\dots,u_h$ say. Each vertex u_i lies in one of the s-sets V_j^s of R_s, which defines a map $\sigma\colon i \mapsto j$. Our aim is to define an embedding $u_i \mapsto v_i \in V_{\sigma(i)}$ of H in G as a subgraph; thus, $v_1,\dots,v_h$ will be distinct, and v_iv_j will be an edge of G whenever u_iu_j is an edge of H.

Our plan is to choose the vertices $v_1, \dots, v_h$ inductively. Throughout the induction, we shall have a 'target set' $Y_i \subseteq V_{\sigma(i)}$ assigned to each u_i; this contains the vertices that are still candidates for the choice of v_i. Initially, Y_i is the entire set $V_{\sigma(i)}$. As the embedding proceeds, Y_i will get smaller and smaller (until it collapses to $\{v_i\}$ when v_i is chosen): whenever we choose a vertex v_j with $j < i$ and $u_j u_i \in E(H)$, we delete all those vertices from Y_i that are not adjacent to v_j. The set Y_i thus evolves as

$$V_{\sigma(i)} = Y_i^0 \supseteq \dots \supseteq Y_i^i = \{v_i\},$$

where Y_i^j denotes the version of Y_i current after the definition of v_j and the resulting deletion of vertices from Y_i^{j-1}.

In order to make this approach work, we have to ensure that the target sets Y_i do not get too small. When we come to embed a vertex u_j, we consider all the indices $i > j$ with $u_j u_i \in E(H)$; there are at most Δ such i. For each of these i, we wish to select v_j so that

$$Y_i^j = N(v_j) \cap Y_i^{j-1} \tag{2}$$

is still relatively large: smaller than Y_i^{j-1} by no more than a constant factor such as $(d - \epsilon)$. Now this can be done by Lemma 7.5.1 (with $A = V_{\sigma(j)}$, $B = V_{\sigma(i)}$ and $Y = Y_i^{j-1}$): provided that Y_i^{j-1} still has size at least $\epsilon\ell$ (which induction will ensure), all but at most $\epsilon\ell$ choices of v_j will be such that the new set Y_i^j as in (2) satisfies

$$|Y_i^j| \geqslant (d-\epsilon)|Y_i^{j-1}|. \tag{3}$$

Excluding the bad choices for v_j for all the relevant values of i simultaneously, we find that all but at most $\Delta\epsilon\ell$ choices of v_j from $V_{\sigma(j)}$, and in particular from $Y_j^{j-1} \subseteq V_{\sigma(j)}$, satisfy (3) for all i.

It remains to show that the sets Y_i^{j-1} considered above as Y for Lemma 7.5.1 never fall below the size of $\epsilon\ell$, and that when we come to select $v_j \in Y_j^{j-1}$ we have a choice of at least s suitable candidates: since before u_j at most $s-1$ vertices u were given an image in $V_{\sigma(j)}$, we can then choose v_j distinct from these.

But all this follows from our choice of ϵ_0. Indeed, the initial target sets Y_i^0 have size ℓ, and each Y_i shrinks at most Δ times by a factor of $(d-\epsilon)$ when some v_j with $j < i$ and $u_j u_i \in E(H)$ is defined. Thus,

$$|Y_i^{j-1}| - \Delta\epsilon\ell \underset{(3)}{\geqslant} (d-\epsilon)^\Delta \ell - \Delta\epsilon\ell \geqslant (d-\epsilon_0)^\Delta \ell - \Delta\epsilon_0\ell \underset{(1)}{\geqslant} \tfrac{1}{2}d^\Delta\ell \geqslant s$$

for all $j \leqslant i$; in particular, we have $|Y_i^{j-1}| \geqslant \epsilon\ell$ and $|Y_j^{j-1}| - \Delta\epsilon\ell \geqslant s$ as desired. □

We are now ready to prove the Erdős-Stone theorem.

(7.1.1) (7.1.4) (7.4.1) r, s γ

Proof of Theorem 7.1.2. Let $r \geqslant 2$ and $s \geqslant 1$ be given as in the statement of the theorem. For $s = 1$ the assertion follows from Turán's theorem, so we assume that $s \geqslant 2$. Let $\gamma > 0$ be given; this γ will play the role of the ϵ of the theorem. If any graph G with $|G| =: n$ has

$\|G\|$

$$\|G\| \geqslant t_{r-1}(n) + \gamma n^2$$

edges, then $\gamma < 1$. We want to show that $K^r_s \subseteq G$ if n is large enough.

Our plan is to use the regularity lemma to show that G has a regularity graph R dense enough to contain a K^r by Turán's theorem. Then R_s contains a K^r_s, so we may hope to use Lemma 7.5.2 to deduce that $K^r_s \subseteq G$.

d, Δ, ϵ_0 On input $d := \gamma$ and $\Delta := \Delta(K^r_s)$ Lemma 7.5.2 returns an $\epsilon_0 > 0$.

m, ϵ To apply the regularity lemma, let $m > 1/\gamma$ and choose $\epsilon > 0$ small enough that $\epsilon \leqslant \epsilon_0$,

$$\epsilon < \gamma/2 < 1\,, \tag{1}$$

and

δ

$$\delta := 2\gamma - \epsilon^2 - 4\epsilon - d - \frac{1}{m} > 0\,;$$

this is possible, since $2\gamma - d - \frac{1}{m} > 0$. On input ϵ and m, the regularity

M lemma returns an integer M. Let us assume that

n

$$n \geqslant \frac{2Ms}{d^\Delta(1-\epsilon)}\,.$$

Since this number is at least m, the regularity lemma provides us with

k an ϵ-regular partition $\{V_0, V_1, \ldots, V_k\}$ of G, where $m \leqslant k \leqslant M$; let

ℓ $|V_1| = \ldots = |V_k| =: \ell$. Then

$$n \geqslant k\ell\,, \tag{2}$$

and

$$\ell = \frac{n - |V_0|}{k} \geqslant \frac{n - \epsilon n}{M} = n\,\frac{1-\epsilon}{M} \geqslant \frac{2s}{d^\Delta}$$

R by the choice of n. Let R be the regularity graph of G with parameters ϵ, ℓ, d corresponding to the above partition. Then Lemma 7.5.2 will imply $K^r_s \subseteq G$ as desired if $K^r \subseteq R$ (and hence $K^r_s \subseteq R_s$).

Our plan was to show $K^r \subseteq R$ by Turán's theorem. We thus have to check that R has enough edges, i.e. that enough ϵ-regular pairs (V_i, V_j) have density at least d. This should follow from our assumption that G has at least $t_{r-1}(n) + \gamma n^2$ edges, i.e. an edge density of about $\frac{r-2}{r-1} + 2\gamma$: this lies substantially above the approximate density of $\frac{r-2}{r-1}$ of the Turán graph $T^{r-1}(k)$, and hence substantially above any density that G could derive from $t_{r-1}(k)$ dense pairs alone, even if all these had density 1.

Let us then estimate $\|R\|$ more precisely. How many edges of G lie outside ϵ-regular pairs? At most $\binom{|V_0|}{2}$ edges lie inside V_0, and by condition (i) in the definition of ϵ-regularity these are at most $\frac{1}{2}(\epsilon n)^2$ edges. At most $|V_0|k\ell \leqslant \epsilon n^2$ edges join V_0 to other partition sets. The at most ϵk^2 other pairs (V_i, V_j) that are not ϵ-regular contain at most ℓ^2 edges each, together at most $\epsilon k^2 \ell^2$. The ϵ-regular pairs of insufficient density ($< d$) each contain no more than $d\ell^2$ edges, altogether at most $\frac{1}{2}k^2 d\ell^2$ edges. Finally, there are at most $\binom{\ell}{2}$ edges inside each of the partition sets $V_1, \dots, V_k$, together at most $\frac{1}{2}\ell^2 k$ edges. All *other* edges of G lie in ϵ-regular pairs of density at least d, and thus contribute to edges of R. Since each edge of R corresponds to at most ℓ^2 edges of G, we thus have in total

$$\|G\| \leqslant \tfrac{1}{2}\epsilon^2 n^2 + \epsilon n^2 + \epsilon k^2 \ell^2 + \tfrac{1}{2}k^2 d\ell^2 + \tfrac{1}{2}\ell^2 k + \|R\|\,\ell^2.$$

Hence, for all sufficiently large n,

$$\begin{aligned}
\|R\| &\geqslant \tfrac{1}{2}k^2\,\frac{\|G\| - \frac{1}{2}\epsilon^2 n^2 - \epsilon n^2 - \epsilon k^2\ell^2 - \frac{1}{2}dk^2\ell^2 - \frac{1}{2}k\ell^2}{\frac{1}{2}k^2\ell^2} \\
&\underset{(1,2)}{\geqslant} \tfrac{1}{2}k^2\left(\frac{t_{r-1}(n) + \gamma n^2 - \frac{1}{2}\epsilon^2 n^2 - \epsilon n^2}{n^2/2} - 2\epsilon - d - \frac{1}{k}\right) \\
&\geqslant \tfrac{1}{2}k^2\left(\frac{t_{r-1}(n)}{n^2/2} + 2\gamma - \epsilon^2 - 4\epsilon - d - \frac{1}{m}\right) \\
&= \tfrac{1}{2}k^2\left(t_{r-1}(n)\binom{n}{2}^{-1}\left(1 - \frac{1}{n}\right) + \delta\right) \\
&> \tfrac{1}{2}k^2\,\frac{r-2}{r-1} \\
&\geqslant t_{r-1}(k)\,.
\end{aligned}$$

(The strict inequality follows from Lemma 7.1.4.) Therefore $K^r \subseteq R$ by Theorem 7.1.1, as desired. □

Having seen a typical application of the regularity lemma in full detail, let us now step back and try to separate the wheat from the chaff: what were the main ideas, how do the various parameters depend on each other, and in which order were they chosen?

The task was to show that γn^2 more edges than can be accommodated on n vertices without creating a K^r force a K^r_s subgraph, provided that G is large enough. The plan was to do this using Lemma 7.5.2, which asks for the input of two parameters: d and Δ. As we wish to find a copy of $H = K^r_s$ in G, it is clear that we must choose $\Delta := \Delta(K^r_s)$. We shall return to the question of how to choose d in a moment.

Given d and Δ, Lemma 7.5.2 tells us how small we must choose ϵ to make the regularity lemma provide us with a suitable partition. The regularity lemma also requires the input of a lower bound m for the number of partition classes; we shall discuss this below, together with d.

All that remains now is to choose G large enough that the partition classes have size at least $2s/d^\Delta$, as required by Lemma 7.5.2. (The s here depends on the graph H we wish to embed, and $s := |H|$ would certainly be big enough. In our case, we can use the s from our $H = K^r_s$.) How large is 'large enough' for $|G|$ follows straight from the upper bound M on the number of partition classes returned by the regularity lemma: roughly, i.e. disregarding V_0, an assumption of $|G| \geqslant 2Ms/d^\Delta$ suffices.

So far, everything was entirely straightforward, and standard for any application of the regularity lemma of this kind. But now comes the interesting bit, the part specific to this proof: the observation that, if only d is small enough, our γn^2 'additional edges' force an 'additional dense ϵ-regular pair' of partition sets, giving us more than $t_{r-1}(k)$ dense ϵ-regular pairs in total (where 'dense' means 'of density at least d'), thus forcing R to contain a K^r and hence R_s to contain a K^r_s.

Let us examine why this is so. Suppose we have at most $t_{r-1}(k)$ dense ϵ-regular pairs . Inside these, G has at most

$$\tfrac{1}{2}k^2\frac{r-2}{r-1}\,\ell^2 \leqslant \tfrac{1}{2}n^2\frac{r-2}{r-1}$$

edges, even if we use those pairs to their full capacity of ℓ^2 edges each (where ℓ is again the common size of the partition sets other than V_0, so that $k\ell$ is nearly n). Thus, we have almost exactly our γn^2 additional edges left to accommodate elsewhere in the graph: either in ϵ-regular pairs of density less than d, or in some exceptional way, i.e. in irregular pairs, inside a partition set, or with an end in V_0. Now the number of edges in low-density ϵ-regular pairs is less than

$$\tfrac{1}{2}k^2 d\ell^2 \leqslant \tfrac{1}{2}dn^2,$$

and hence less than half of our extra edges if $d \leqslant \gamma$. The other half, the remaining $\frac{1}{2}\gamma n^2$ edges, are more than can be accommodated in exceptional ways, provided we choose m large enough and ϵ small enough (giving an additional upper bound for ϵ). It is now a routine matter to compute the values of m and ϵ that will work.

Exercises

1.$^-$ Show that $K_{1,3}$ is extremal without a P^3.

2.$^-$ Given $k > 0$, determine the extremal graphs of chromatic number at most k.

3.$^-$ Is there a graph that is edge-maximal without a K^3 minor but not extremal?

4. Determine the value of $\mathrm{ex}(n, K_{1,r})$ for all $r, n \in \mathbb{N}$.

5.$^+$ Given $k > 0$, determine the extremal graphs without a matching of size k.

(Hint. Theorem 2.2.3 and Ex. 20, Ch. 2.)

6. Without using Turán's theorem, show that the maximum number of edges in a triangle-free graph of order $n > 1$ is $\lfloor n^2/4 \rfloor$.

7. Show that

$$t_{r-1}(n) \leqslant \tfrac{1}{2} n^2 \, \frac{r-2}{r-1} \,,$$

with equality whenever $r - 1$ divides n.

8. Show that $t_{r-1}(n)/\binom{n}{2}$ converges to $(r-2)/(r-1)$ as $n \to \infty$.

(Hint. $t_{r-1}\big((r-1)\lfloor \frac{n}{r-1} \rfloor\big) \leqslant t_{r-1}(n) \leqslant t_{r-1}\big((r-1)\lceil \frac{n}{r-1} \rceil\big)$.)

9. Does every large enough graph G with at most $c\,|G|$ edges, where c is any constant, contain a set of 100 independent vertices?

10. Show that deleting at most $(m-s)(n-t)/s$ edges from a $K_{m,n}$ will never destroy all its $K_{s,t}$ subgraphs.

11. For $0 < s \leqslant t \leqslant n$ let $z(n, s, t)$ denote the maximum number of edges in a bipartite graph whose partition sets both have size n, and which does not contain a $K_{s,t}$. Show that $2\,\mathrm{ex}(n, K_{s,t}) \leqslant z(n, s, t) \leqslant \mathrm{ex}(2n, K_{s,t})$.

12.$^+$ Let $1 \leqslant r \leqslant n$ be integers. Let G be a bipartite graph with bipartition $\{A, B\}$, where $|A| = |B| = n$, and assume that $K_{r,r} \not\subseteq G$. Show that

$$\sum_{x \in A} \binom{d(x)}{r} \leqslant (r-1)\binom{n}{r}.$$

Using the previous exercise, deduce that $\mathrm{ex}(n, K_{r,r}) \leqslant cn^{2-1/r}$ for some constant c depending only on r.

13. The *upper density* of an infinite graph G is the lim sup of the maximum edge densities of its (finite) n-vertex subgraphs as $n \to \infty$.

   (i) Show that, for every $r \in \mathbb{N}$, every infinite graph of upper density $> \frac{r-2}{r-1}$ has a K^r_s subgraph for every $s \in \mathbb{N}$.

   (ii) Deduce that the upper density of infinite graphs can only take the countably many values of $0, 1, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$.

14. Given a tree T, find an upper bound for $\mathrm{ex}(n, T)$ that is linear in n and independent of the structure of T, i.e. depends only on $|T|$.

15. Show that the Erdős-Sós conjecture is best possible in the sense that, for every k and infinitely many n, there is a graph on n vertices and with $\frac{1}{2}(k-1)n$ edges that contains no tree with k edges.

16.$^-$ Prove the Erdős-Sós conjecture for the case when the tree considered is a star.

17. Prove the Erdős-Sós conjecture for the case when the tree considered is a path.

(Hint. Use Exercise 9 of Chapter 1.)

18. Can large average degree force the chromatic number up if we exclude some tree as an induced subgraph? More precisely: For which trees T is there a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, for every $k \in \mathbb{N}$, every graph of average degree at least $f(k)$ either has chromatic number at least k or contains an induced copy of T?

19. Given two numerical graph invariants i_1 and i_2, write $i_1 \leqslant i_2$ if we can force i_2 to be arbitrarily high on some subgraph of G by assuming that $i_1(G)$ is large enough. (Formally: write $i_1 \leqslant i_2$ if there exists a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, given any $k \in \mathbb{N}$, every graph G with $i_1(G) \geqslant f(k)$ has a subgraph H with $i_2(H) \geqslant k$.) If $i_1 \leqslant i_2$ as well as $i_1 \geqslant i_2$, write $i_1 \sim i_2$. Show that this is an equivalence relation for graph invariants, and sort the following invariants into equivalence classes ordered by $<$: minimum degree; average degree; connectivity; arboricity; chromatic number; colouring number; choice number; $\max\{\, r \mid K^r \subseteq G \,\}$; $\max\{\, r \mid TK^r \subseteq G \,\}$; $\max\{\, r \mid K^r \preccurlyeq G \,\}$; $\min \max d^+(v)$, where the maximum is taken over all vertices v of the graph, and the minimum over all its orientations.

20.$^+$ Prove, from first principles and without using average or minimum degree arguments, the existence of a function $f\colon \mathbb{N} \to \mathbb{N}$ such that every graph of chromatic number at least $f(r)$ has a K^r minor.

(Hint. Use induction on r. For the induction step $(r-1) \to r$ try to find a connected set U of vertices whose neighbours induce a subgraph that needs enough colours to contract to K^{r-1}. If no such set U exists, show that the given graph can be coloured with fewer colours than assumed.)

21. Given a graph G with $\varepsilon(G) \geqslant k \in \mathbb{N}$, find a minor $H \preccurlyeq G$ such that $\delta(H) \geqslant k \geqslant |H|/2$.

22.$^+$ Find a constant c such that every graph with n vertices and at least $n + 2k(\log k + \log\log k + c)$ edges contains k edge-disjoint cycles (for all $k \in \mathbb{N}$). Deduce an edge-analogue of the Erdős-Pósa theorem (2.3.2).

(Hint. Assuming $\delta \geqslant 3$, delete the edges of a short cycle and apply induction. The calculations are similar to the proof of Lemma 2.3.1.)

23. Simplify the proof of Theorem 7.2.3 by using Exercise 32 of Chapter 3.

24.$^+$ Show that any function h as in Lemma 3.5.1 satisfies the inequality $h(r) > \frac{1}{8}r^2$ for all even r, and hence that Theorem 7.2.3 is best possible up to the value of the constant c.

25. Characterize the graphs with n vertices and more than $3n-6$ edges that contain no $TK_{3,3}$. In particular, determine $\mathrm{ex}(n, TK_{3,3})$.

(Hint. You may use the theorem of Wagner that every edge-maximal graph without a $K_{3,3}$ minor can be constructed recursively from maximal planar graphs and copies of K^5 by pasting along K^2s.)

26.$^-$ Derive the four colour theorem from Hadwiger's conjecture for $r = 5$.

27.$^-$ Show that Hadwiger's conjecture for $r+1$ implies the conjecture for r.

28.$^-$ Deduce the following weakening of Hadwiger's conjecture from known results: given any $\epsilon > 0$, every graph of chromatic number at least $r^{1+\epsilon}$ has a K^r minor, provided that r is large enough.

29.$^-$ Show that any graph constructed as in Proposition 7.3.1 is edge-maximal without a K^4 minor.

30. Prove the implication $\delta(G) \geqslant 3 \Rightarrow G \supseteq TK^4$.

(Hint. You may use any result from Section 7.3.)

31. A multigraph is called *series-parallel* if it can be constructed recursively from a K^2 by the operations of subdividing and of doubling edges. Show that a 2-connected multigraph is series-parallel if and only if it has no (topological) K^4 minor.

32. Without using Theorem 7.3.8, prove Hadwiger's conjecture for all graphs of girth at least 11 and r large enough. Without using Corollary 7.3.9, show that there is a constant $g \in \mathbb{N}$ such that all graphs of girth at least g satisfy Hadwiger's conjecture, irrespective of r.

33.$^+$ Prove Hadwiger's conjecture for $r = 4$ from first principles.

34.$^+$ Prove Hadwiger's conjecture for line graphs.

35. Prove Corollary 7.3.5.

36.$^-$ In the definition of an ϵ-regular pair, what is the purpose of the requirement that $|X| \geqslant \epsilon |A|$ and $|Y| \geqslant \epsilon |B|$?

37.$^-$ Show that any ϵ-regular pair in G is also ϵ-regular in $\overline{G}$.

38. Consider a partition of a finite set V into k equally sized subsets. Show that the complete graph on V has about $k-1$ as many edges between different partition sets as edges inside partition sets. Explain how this leads to the choice of $m := 1/\gamma$ in the proof of the Erdős-Stone theorem.

39. (i) Deduce the regularity lemma from the assumption that it holds, given $\epsilon > 0$ and $m \geqslant 1$, for all graphs of order at least some $n = n(\epsilon, m)$.

(ii) Prove the regularity lemma for sparse graphs—more precisely, for every sequence $(G_n)_{n\in\mathbb{N}}$ of graphs G_n of order n such that $\|G_n\|/n^2 \to 0$ as $n \to \infty$.

Notes

The standard reference work for results and open problems in extremal graph theory (in a very broad sense) is still B. Bollobás, *Extremal Graph Theory*, Academic Press 1978. A kind of update on the book is given by its author in his chapter of the *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995. An instructive survey of extremal graph theory in the narrower sense of Section 7.1 is given by M. Simonovits in (L.W. Beineke & R.J. Wilson, eds.) *Selected Topics in Graph Theory 2*, Academic Press 1983. This paper focuses among other things on the particular role played by the Turán graphs. A more recent survey by the same author can be found in (R.L. Graham & J. Nešetřil, eds.) *The Mathematics of Paul Erdős*, Vol. 2, Springer 1996.

Turán's theorem is not merely one extremal result among others: it is the result that sparked off the entire line of research. Our first proof of Turán's theorem is essentially the original one; the second is a version of a proof of Zykov due to Brandt.

Túran's theorem has been generalized as follows. Suppose that, for some fixed $r \geqslant 3$, we wish to construct a graph on n vertices with at least γn^2 edges, where now $\frac{1}{2}\frac{r-2}{r-1} < \gamma < \frac{1}{2}$, in such a way as to create as few K^r subgraphs as possible. The *clique density theorem* says that, for fixed γ, the asymptotically best way to do this is to form a complete multipartite graph in which all classes have the same size except for one, which may be smaller. How many such classes there are depends on γ, but not on n: as in Turán's theorem, s classes will always give about γn^2 edges for $\gamma = \frac{1}{2}\frac{s-1}{s}$. The clique density theorem had been conjectured by Lovász and Simonovits in 1983, and was finally proved for all r by C. Reiher, The clique density theorem, *Ann. Math.* **184** (2016), 683–707, arXiv:1212.2454.

Our version of the Erdős-Stone theorem is a slight simplification of the original. A direct proof, not using the regularity lemma, is given in L. Lovász, *Combinatorial Problems and Exercises* (2nd edn.), North-Holland 1993. Its most fundamental application, Corollary 7.1.3, was only found 20 years after the theorem, by Erdős and Simonovits (1966).

Of our two bounds on $\mathrm{ex}(n, K_{r,r})$ the upper one is thought to give the correct order of magnitude. For vastly off-diagonal complete bipartite graphs this was verified by J. Kollár, L. Rónyai & T. Szabó, Norm-graphs and bipartite Turán numbers, *Combinatorica* **16** (1996), 399–406, who proved that $\mathrm{ex}(n, K_{r,s}) \geqslant c_r n^{2-\frac{1}{r}}$ when $s > r!$.

Details about the Erdős-Sós conjecture, including an approximate solution for large k, can be found in the survey by Komlós and Simonovits cited below. The case where the tree T is a path (Exercise 17) was proved by Erdős & Gallai in 1959. It was this result, together with the easy case of stars (Exercise 16) at the other extreme, that inspired the conjecture as a possible unifying result. A proof of the precise conjecture for large graphs was announced in 2009 by Ajtai, Komlós, Simonovits and Szemerédi, but has not been made publicly available.

The Erdős-Sós conjecture says that graphs of average degree greater than $k-1$ contain every tree with k edges. Loebl, Komlós and Sós have conjectured a 'median' version, which appears to be easier: that if at least half the vertices